

EwERunConsole

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Scope

- The EwERunConsole provides a flexible and customizable framework to run EwE simulations through code
- Through the console, EwE executions are controlled via text files. This means that EwE can be controlled via any programming language.

Concept

- The EwE models are wrapped in a console application.
- The console application is operating-system agnostic (.NET 8 core).
- The console application, with all its dependencies, is provided in a single folder.
- The console application takes two parameters:
 1. A configuration file, in [Json format](#), that contains EwE run instructions.
 2. The output folder where simulation results will be placed.
- The configuration file is setup to allow users to vary almost any parameter in the Ecopath, Ecosim and Ecospace models. Data can be obtained from accompanying files, too,
- Users can now use code to write Json configuration files to disk, perturb and execute the EwE models, and analyse EwE output, without having to know .NET.

Example

- Extract the contents of the ZIP file “EwERunConsole” to a folder on your machine. This will provide
- Open a command prompt, and navigate to the folder “ewe”
- Copy and paste the following in the command prompt and press enter:

```
EwERunConsole.exe --i "AnchovyBay_runinfo.json" --o "AnchovyBay_out"
```

The parameters are:

- --i: the absolute or relative path to the Json file.
- --o: the absolute or relative path to a folder to store the output.

What happens in response:

- The console will read the instructions provided in the .Json file to configure the EwE models and execute them.
- Output will be aggregated in the indicated output folder.
- The console will verbosely explain the steps it has taken to facilitate troubleshooting. This output is also written to a file “EwERunConsole_log.txt” in the output folder.

The configuration file

The configuration file is a JSON file, which is a simple, text-based format with a few basic rules ([info here](#)). EII has defined a JSON format to automate (e.g, control the configuration and execution) of Ecopath, Ecosim and Ecospace.

```
{
  "Configuration": {
    "ModelFile": "AnchovyBay.eiixml",
    "EcosimScenario": 1,
    "EcosimTimeseries": 0,
    "EcospaceScenario": 1,
    "SaveWithHeader": true,
    "ExtDataConfigFile": "AnchovyBay_STDF.xml",
    "RunYears": 50
  },
  "EcospaceRun": {
    "RunSpinupYears": 10,
    "UseIBM": false,
    "MaxCores": 4,
    "MinHabCap": 0.001,
    "SaveFirstYear": 2005,
    "SaveAnnual": true,
    "SaveContentAsc": [ "biomass", "catch", "effort" ],
    "SaveContentCsv": [ "xyz" ],
    "Changes": [
      {
        "Date": "start",
        "Modifications": {
          "ecospace.mpa[1].map.fill": 0
        }
      },
      {
        "Date": "1-1-2025",
        "Modifications": {
          "ecospace.mpa[1].map.load": "mpa_test.asc"
        }
      }
    ]
  }
}
```

Code snippet 2. The JSON file used to execute the Anchovy Bay model with the spatial temporal data framework. Ecospace uses a 10-year spin-up period, saves ASCII maps of Biomass, Catch and effort, and regional averages CSV files. While running, the various changes come into force to alter the map layout of MPA 1. The map is cleared of any remaining cells at the start of the simulation. At the time step corresponding to 1-1-2025 the MPA receives a new cell layout from a file.

The EwE automation syntax is much elaborate than the brief example shown here, but this provides the basic idea.

Note that the EwERunConsole will report on any JSON element that it cannot recognize. Only when the entire JSON file can be interpreted, EwERunConsole will run the EwE models.

.JSON file structure and fields explained

Field	Meaning	Must be specified?
{	Opening bracket for JSON text	Yes
"Configuration": {	Start of "Configuration" block to configure generic EwE parameters	Yes
"ModelFile": "",	Specify which EwE model to load, in .eiixml format (exported from EwE, see next section)	Yes
"EcosimScenario": 1,	The one-based Ecosim scenario to load. Omit or set 0 or less to not load Ecosim (and not load Ecospace too, of course)	No
"EcosimTimeseries": 1,	The one-based Ecosim time series dataset to load	No
"EcospaceScenario": 1,	The one-based Ecospace scenario to load	No
"SaveWithHeader": true,	If true, any EwE model output is saved to disk with a header file with contextual information.	No
"ExtDataConfigFile": "",	Path to the STDF configuration file (exported from EwE, see next section)	No
"RunYears": 50	The number of years to run the indicated Ecosim and Ecospace scenarios for. If omitted, the default run length of the model is used.	No
},	End of "Configuration" block	Yes
"EcospaceRun": {	Start of "Ecospace" block to configure static Ecospace parameters and which changes to make while Ecospace runs	No
"RunSpinupYears": 10,	Number of years to run the spin-up for. If omitted, the spin-up is not used	No
"UseIBM": false,	Run Ecospace in IBM mode (true) or MultiStanza (false). If omitted, MultiStanza is used	No
"MaxCores": 4,	The number of threads to allocate to Ecospace	No
"MinHabCap": 0.001,	If non-zero, every cell in Ecospace will at least have the foraging capacity indicated here. If omitted, the default of 0.01 is used. If set to 0, cells can have 0 habitat foraging capacity	No
"SaveFirstYear": 2005,	Set a custom year to start saving output for	No
"SaveAnnual": true,	Save annual (true) or monthly (false) data to file. Default is monthly	No
"SaveContentAsc": [...],	Indicates which Ecospace maps to save to disk	No
"SaveContentCsv": [...],	Indicates which Ecospace CSV data to save to disk	No
"Changes": [	Start of an array of changes	No
{	Start of the a change set	Yes
"Date": "start",	The date to make the change(s). You can also specify "Timestep": 1 instead of "Date": "start"	Yes
"Modifications": {	Start of the modifications to make at this time step	Yes
"ecospace.mpa[1].map.fill": 0, "ecospace.mpa[2].map.fill": 0	A collection of "action": "value" statements to make changes in Ecopath, Ecosim and Ecospace while the model runs. You can have as many "action": "value" pairs as you want.	
}	End of a modifications	Yes
},	End of change set	Yes
{	Next change set... etc	
"Date": "1-1-2025",		
"Modifications": {		
"ecospace.mpa[1].map.load": "mpa_test.asc"		
}		
}		
]	End of array of changes	Must close if used
}	End of "Ecospace" block	Must close if used
}	End of JSON file	Yes

Preparing for running experiments

There are many ways to organize the executable, input data and run JSON files, but it is important that all bits are in place and can be found by the EwERunConsole.

It is recommended to use folder structures similar to:

```
[EwERunConsole]
|- [InputData]
|   |- [Experiment1]
|   |   |- {data files for experiment 1}
|   |
|   |- [Experiment2]
|   |   :
|   |   :
|   |
|   |- [STDFData]
|   |   |- MySTDFData.xml
|   |   |- {data files shared by all experiments}
|   |   |- experiment1.json
|   |   |- experiment2.json
|   |   | :
|   |   | :
|   |   |- MyEwEModel.eiixml
|
|- [OutputData]
|   |- {here the tool can create an output folder for each experiment}
|
|- [Executable]
|   |- EwERunConsole.exe
|   |- {supporting files}
```

To execute the first experiment with the EwERunConsole, one would use the following command:

```
EwERunConsole.exe --i "..\InputData\experiment1.json" --o "..\OutputData"
```

Where:

- Input and Output folders are defined relatively to the folder with the EwERunConsole.exe;
- The EwERunConsole will presume that the folder with the .json file is the working directory containing all other input files that the .json file refers to. For instance, to load the model file, the experiment1.json simply refers to “MyModel.eiixml”. Similarly, experiment1.json can load spatial temporal data by referring to “MySTDFData.xml”.
- To refer to a specific file in the Experiment1 folder, experiment1.json could use the relative path “Experiment1\somefile.asc”

1. The EwE model

To prepare the model for the run console you must export it to XML. The run console will not let you use Access databases; these models can’t be opened on non-Windows operating systems. You can convert a working EwE model to XML in the EwE software:

- Load the model in EwE
- Navigate to EwE > Menu > File > Export to XML...
The XML version of the model will be created as an .eiixml file in the same folder as the original (Access) model.
- Copy the generated .eiixml to the input data package for the run console and must be referred from the JSON configuration file.

Preparing the External Data for the Spatial Temporal Data Framework

The EwERunConsole can use the Spatial Temporal Data Framework (STDF) with some extra preparation.

The STDF uses absolute file paths to identify the time varying maps on a users' computer. However, as EwERunFramework exercises should be easily moved between computers, it is more practical of the framework were to use relative paths instead. You can achieve that from the EwE software too.

To export spatial temporal data defined in an EwE model ([see guide](#)):

- Load the model in EwE
- Load Ecosim and Ecospace too
- Navigate to EwE > Menu > Ecospace > Define external connections...
- Press “Export”. When prompted, also export the data. Pick a folder on your computer and the spatial temporal data framework will create a STDF configuration file and folders with maps underneath that uses only relative file paths
- Copy the STDF configuration file and map data folders to your EwERunConsole data folder and voila.

Future work

- Need to decide how to name and aggregate output.
- The console will need to incorporate control over specific plug-ins such as EcoInd, MultiSim, ENA, etc.
- The console will need to incorporate Ecotracer and built-in searches.
- Add a comprehensive description of the available JSON syntax, which the command line application must be able to write out.